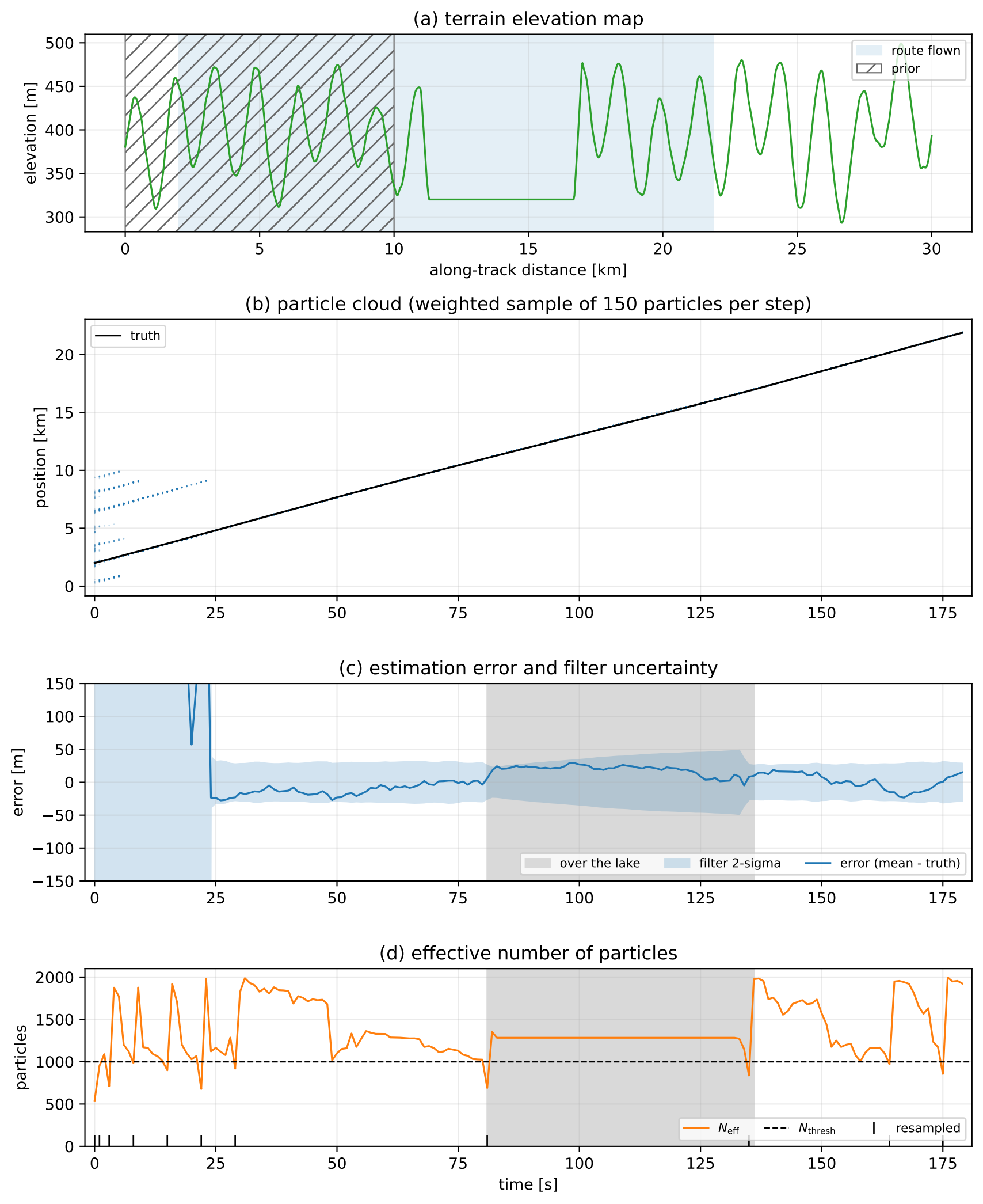

(a) terrain elevation map
elevation [m]
along-track distance [km]
route flown
prior
(b) particle cloud (weighted sample of 150 particles per step)
position [km]
truth
(c) estimation error and filter uncertainty
error [m]
over the lake
filter 2-sigma
error (mean - truth)
(d) effective number of particles
particles
time [s]
N_{eff}
N_{thresh}
resampled